

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CT
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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

17 September 2018

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Section A:

Answer **ALL** questions in the spaces provided.

Section B:

Answer any **ONE** question.
Write your answers on the separate writing papers provided.

Please hand in section A and section B separately.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 30
2	/ 10
3	/ 10
Section B	
4 or 5	/ 25
Total	/ 75

Section A

Answer **all** questions in this section.

- 1 Fig. 1.1 shows a molecule of haemoglobin. The haem group plays an important role in the function of haemoglobin.

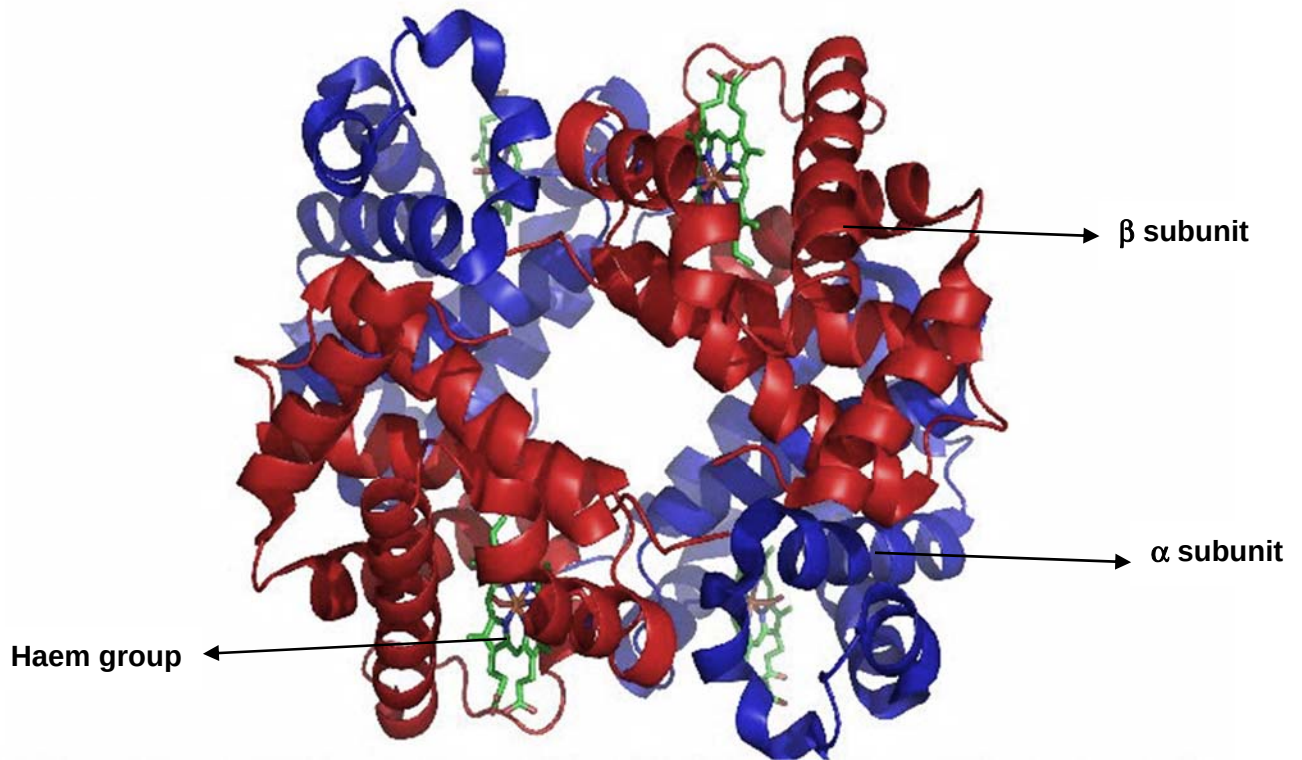


Fig. 1.1

- (a) With reference to Fig. 1.1 and your knowledge, describe **two** structural differences between tropocollagen and haemoglobin.

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..... [2]

- (b)** Discuss the advantages of having four subunits in haemoglobin.

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..... [3]

- (c)** Myoglobin is a protein found in the muscle cells of animals. It functions as an oxygen-storage unit, providing oxygen to the working muscles.

Fig. 1.2 shows a molecule of myoglobin.



Fig 1.2

[Turn over

Fig. 1.3 shows the oxygen dissociation curves for myoglobin, fetal haemoglobin and adult haemoglobin.

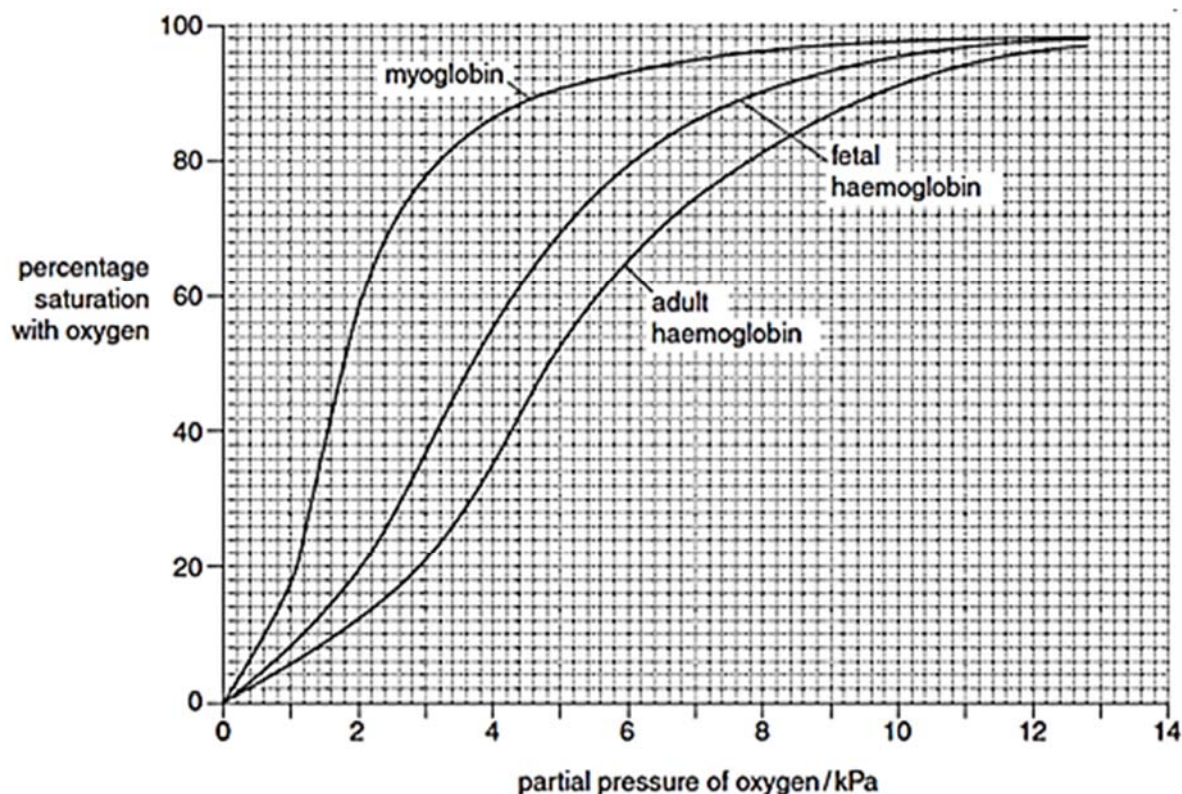


Fig. 1.3

- (i) Name the cells in which haemoglobin is found.

..... [1]

- (ii) Use Fig. 1.3 to determine the percentage saturation of myoglobin and adult haemoglobin when the partial pressure of oxygen is 3 kPa.

..... [1]

- (iii) There is a large difference between the percentage saturation of myoglobin and that of adult haemoglobin at low partial pressures of oxygen. Suggest reasons for this.

..... [2]

- (d) Fetal haemoglobin has a different oxygen binding affinity to that of adult haemoglobin, as shown in Fig. 1.3. Normally, after birth, the production of fetal haemoglobin stops and the adult form is produced.

In a rare condition known as Hereditary Persistence of Fetal Haemoglobin (HPFH), fetal haemoglobin continues to be produced well into adulthood in addition to adult haemoglobin. This condition, however, usually lacks any symptoms.

- (i) Explain, with reference to Fig. 1.3, the significance of the difference in oxygen binding affinity between fetal and adult haemoglobin.

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 [2]

- (ii) Suggest why HPFH usually lacks symptoms.

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 [1]

Haemoglobin disorders are inherited blood diseases that affect the quality or amount of haemoglobin and the capacity to carry oxygen around the body. They fall into two main categories: thalassemia and sickle cell disease.

β -thalassemia is caused by mutations in the β -globin gene. People with severe thalassemia depend on blood transfusions to give them working red blood cells. A type of bone marrow transplant has been used to try to cure thalassemia, but it is hard to find compatible donors. Even if a donor can be found, there is still a risk that the patient's body will reject the transplant.

- (e) Explain why a bone marrow transplant from a healthy donor can be used to cure thalassemia.

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 [2]

[Turn over

- (f) β -thalassaemia major is a severe form of β -thalassaemia requiring lifelong blood transfusion. Cardiac complications are the major cause of death in patients with β -thalassaemia major. Studies have shown that a non-competitive enzyme inhibitor improves cardiac functions in blood transfusion-dependent patients with β -thalassaemia major.

(i) Explain the effects of a non-competitive inhibitor on the rate of enzyme activity.

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..... [3]

Fig. 1.4 shows two ways enzymes interact with their substrates.

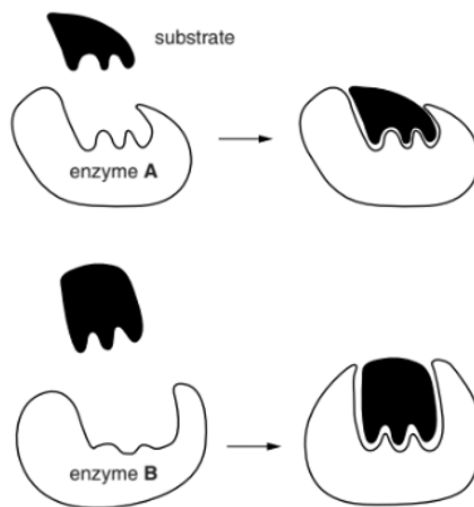


Fig. 1.4

- (ii) Explain the difference between the **two** ways in which enzymes interact with their substrates as shown in Fig. 1.4.

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..... [2]

Sickle cell anemia is an autosomal, recessive human disease. A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in Table 1.1.

Table 1.1

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

(g) Based on the information in Table 1.1,

(i) which patient(s) has / have the sickle cell anemia mutation?

..... [1]

(ii) explain how the change of amino acid would result in a sickle shaped red blood cell.

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..... [3]

[Turn over

Fig. 1.5 shows the distribution of malaria and the allele frequency of the sickle cell allele in Africa.

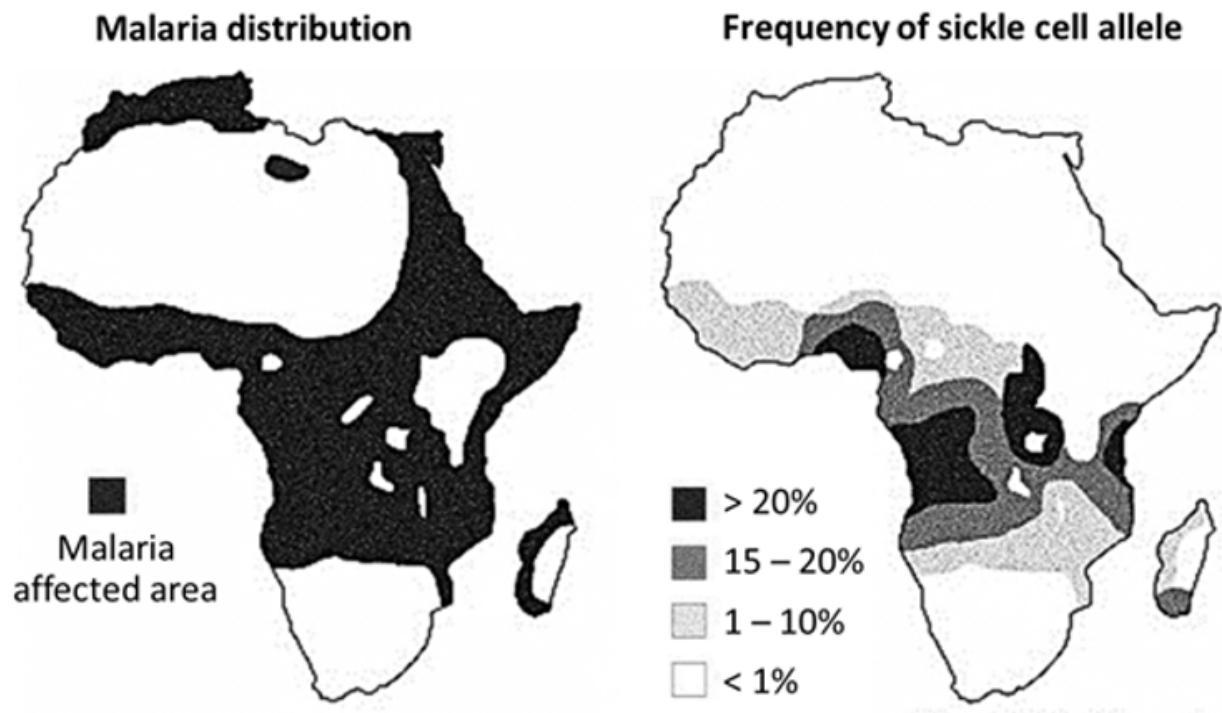


Fig. 1.5

- (h) Explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele.

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..... [3]

The genotypes of five different individuals living in Central Africa are identified by Restriction Fragment Length Polymorphism (RFLP) analysis, using the restriction enzyme *MstII*. Individuals may have the normal allele for the β -globin gene, the sickle cell allele or both alleles.

The β -globin gene sequence in humans is flanked by two *MstII* restriction sites, and in the normal allele, there is another *MstII* restriction site within the gene sequence.

The autoradiograph obtained from the RFLP analysis is shown in Fig. 1.6.

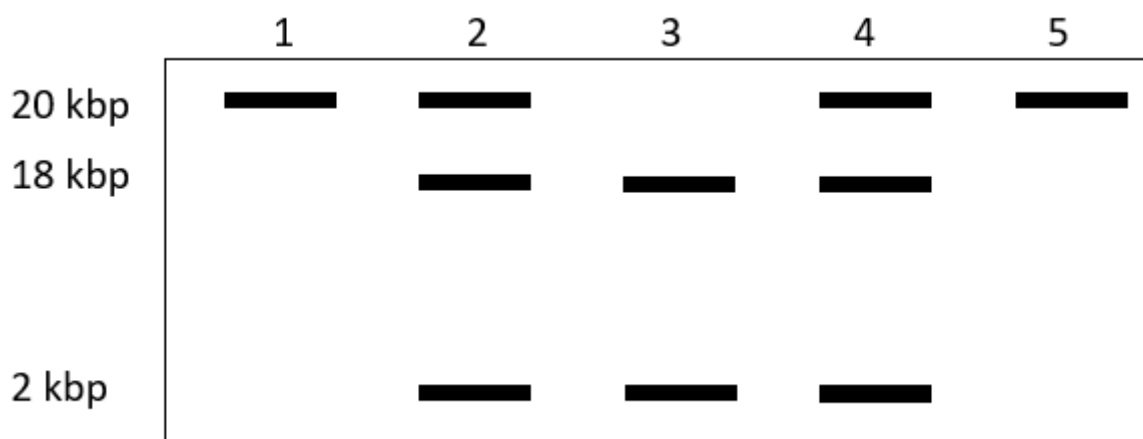


Fig. 1.6

- (i) With reference to Fig. 1.6, identify and explain which individuals are carriers of the sickle cell disease.

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..... [4]

[Total: 30]

[Turn over

- 2 The *lac* operon is involved in the transport and metabolism of lactose in *Escherichia coli*. The operon is regulated by molecules that turn gene expression on and off in response to the concentration of nutrients.

(a) Describe how the concentration of glucose in a bacteria cell regulates the expression of the *lac* operon in the presence of lactose.

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(b) Suggest and explain the importance of operons in bacteria.

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(c) Explain why the mRNA encoded by the *lac* operon is described as polycistronic.

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..... [2]

Fig. 2.1 shows a hybrid operon where the structural genes of the *trp* operon have been fused by deletion to the *lac* operon. The genes necessary for the biosynthesis of the essential amino acid tryptophan remain intact.



Fig. 2.1

(d) Predict the amount of EDCBA proteins by filling the table below using symbols:

“-“ → negligible protein product

“+” → protein product

“++” → lots of protein product

Metabolites present

Glucose	-	+	-	-	+	+	-	+
Lactose	-	-	+	-	-	+	+	+
Tryptophan	-	-	-	+	+	-	+	+
Amount of EDCBA proteins								

[2]

[Total: 10]

[Turn over

- 3 The dengue virus is a member of the genus *Flavivirus* in the family *Flaviviridae*. Along with the dengue virus, this genus also includes a number of other viruses transmitted by mosquitoes and ticks that are responsible for human diseases. *Flavivirus* includes the yellow fever, West Nile, Japanese encephalitis, and tick-borne encephalitis viruses.

The dengue virus is particularly notable in that it exists as four antigenically distinct serotypes (denoted as DENV-1 to DENV-4), within which there is considerable genetic variation in the guise of phylogenetically defined “genotypes”.

- (a) Describe how the dengue virus develops in humans.

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.....[3]

- (b) Describe **two** ways in which the production of antibodies helps in removing DENV from the body in the primary antibody response.

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..... [2]

- (c) Explain why an individual may not be necessarily immunised against a second dengue infection.

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.....[2]

Global warming is the unusually rapid increase in Earth's average surface temperature over the past century primarily due to the greenhouse gases released as a result of anthropogenic activities. The global average surface temperature rose by 0.6 to 0.9°C between 1906 and 2005, and the rate of temperature increase has nearly doubled in the last 50 years. Temperatures are certain to go up further.

(d) Explain how global warming has encouraged the spread of dengue.

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.....[3]

[Total: 10]

[Turn over

Section B

Answer **ONE** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clear labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 **(a)** Describe the production and folding of a functional enzymatic protein that is used within a cell. [13]
- (b)** The effects of global warming on the spread of malaria beyond the tropics is a debatable issue. Discuss the arguments/evidences that support your stance on the matter, and provide a balanced account of counter-arguments. [12]

[Total: 25]

- 5 **(a)** All living organisms need to synthesise ATP. Summarise how cells produce ATP. [13]
- (b)** Discuss why life would be impossible without ATP. [12]

[Total: 25]

END OF PAPER